

Exercise 7: Financial Forecasting

CODE:

```
public class FinancialForecast {  
    public static double futureValue(double presentValue, double growthRate, int years) {  
        // Base case  
        if (years == 0) {  
            return presentValue;  
        }  
        // Recursive case  
        return (1 + growthRate) * futureValue(presentValue, growthRate, years - 1);  
    }  
    public static void main(String[] args) {  
        double presentValue = 10000;  
        double growthRate = 0.10; // 10%  
        int years = 5;  
        double result = futureValue(presentValue, growthRate, years);  
        System.out.printf("Future Value after %d years = %.2f%n", years, result);  
    }  
}
```

OUTPUT:

```
Future Value after 5 years = 16105.10
```